

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

V Semester of B.Sc. Examination, November, 2019

Course code & Name: BC 505 & Human Physiology

Date: 05/11/2019 Day: Tuesday Time: 1.30 PM to 2.00 PM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	The gene for sex determination in mammals is _____ and it is located in _____		01
	(i) MAT, Y.	(ii) SRY, X .	
	(iii) SRY, Y	(iv) MAT, X .	
2.	The hormone erythropoietin stimulates red blood cell production in the red bone marrow. Where in the body is erythropoietin produced?		01
	(i) Spleen	(ii) Kidney.	
	(iii) Liver	(iv) Thyroid	
3.	A membrane-bound structure, at tip of head of a sperm, helping in digesting outer surface of egg, called as:		01
	(i) Sperm.	(ii) Head of sperm.	
	(iii) Acrosome.	(iv) Base of ova.	
4.	The maximum volume of air that can be released from the lungs by forceful expiration after deepest inspiration is called the _____.		01
	(i) Vital capacity.	(ii) Total lung capacity.	
	(iii) Tidal volume.	(iv) Ventilation rate.	
5.	Which malfunction of the respiratory system is the destruction of alveolar walls?		01
	(i) Bronchitis.	(ii) Emphysema.	
	(iii) Pneumonia	(iv) Asthma.	

6.	Write the correct sequence of the pathway through which air travels after entering the body.		01
	(i) Larynx, pharynx, trachea bronchioles.	(ii) Pharynx, larynx, trachea, bronchioles.	
	(iii) Pharynx, larynx, bronchioles, trachea.	(iv) Pharynx, trachea, larynx, bronchioles.	
7.	Where does haematopoiesis take place?		01
	(i) Liver.	(ii) Adipose tissue.	
	(iii) Bone marrow.	(iv) Lungs.	
8.	What would happen to red blood cells if the haem group were removed from haemoglobin?		01
	(i) Red blood cells would not be able to reproduce.	(ii) White blood cells would not be able to reproduce.	
	(iii) Blood clot formation would be inhibited.	(iv) Red blood cells would not be able to bind oxygen.	
9.	Tissue plasmin activator _____		01
	(i) helps in wound healing.	(ii) immunity.	
	(iii) allergy response.	(iv) dissolves clot in blood vessels.	
10.	If your adrenal cortex was producing high levels of aldosterone, it would cause your urine to have		01
	(i) low Na ⁺ and high K ⁺ concentrations	(ii) high Na ⁺ and low K ⁺ concentrations	
	(iii) low Na ⁺ and low K ⁺ concentrations	(iv) high Na ⁺ and high K ⁺ concentrations	
11.	Which of the following function is not carried out by neurons?		01
	(i) the myelination of axons.	(ii) the reuptake of neurotransmitters.	
	(iii) the production of metabotropic receptors.	(iv) the production of microtubules.	
12.	Where do neurons store the genetic information they use to code and build all the proteins required for their functions?		01
	(i) in the axon.	(ii) in the nucleus.	
	(iii) in the synapse	(iv) in the endoplasmic reticulum.	
13.	Electrical impulses gather and accumulate in which part of a neuron, in order to initiate an action potential?		01
	(i) Dendrites	(ii) Axon hillock	
	(iii) Axon terminal branches	(iv) Node of Ranvier	
14.	Most digestion in humans occurs in the first 25 centimeters of the 6-meter length of the small intestine called the _____.		01
	(i) Omasum	(ii) Cecum	
	(iii) Colon	(iv) Duodenum	
15.	James has been having trouble digesting lean steaks and other protein-rich foods, and these foods seem to stay in his stomach longer than usual. James is likely lacking the enzyme ____.		01
	(i) Bile	(ii) Lypase	
	(iii) Amylase	(iv) Pepsin	

16.	Which among the following is the cell division without growth?		01
	(i) Cleavage	(ii) Conjugation	
	(iii) Fertilization	(iv) Development	
17.	If a person undergoes a prolonged fasting then his urine will be found to contain higher levels of		01
	(i) Ketones.	(ii) Fats.	
	(iii) Amino acids.	(iv) Glucose.	
18.	The structure which closes of larynx is _____		01
	(i) Vocal cords.	(ii) Adam's apple.	
	(iii) Glottis.	(iv) Epiglottis.	
19.	How many oxygen molecules bound to hemoglobin to give 50% saturation?		01
	(i) 6	(ii) 4	
	(iii) 2	(iv) 3	
20.	In the absence of acrosome the sperm cannot _____		01
	(i) Penetrate into egg.	(ii) Get food.	
	(iii) Get energy.	(iv) Swim.	

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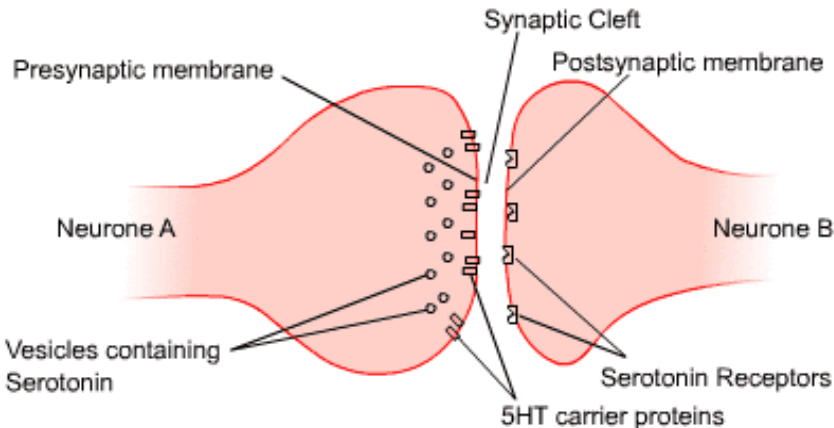
V Semester of B. Sc. Examination, November, 2019

Course code & Name: BC 505 & Human Physiology

Date: 05/11/2019 Day: Tuesday Time: 2.01 PM to 4.30 PM Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEETS.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of a non - programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II Answer the following questions as directed (attempt any 5)		20
1.	What are the three major buffer systems of the body and how do they work?	04
2.	What is Emphysema? What are symptoms and treatments of Emphysema?	04
3.	Compare between internal and external respiration.	04
4.	What is spermatogenesis and Capacitation? Describe briefly.	04
5.	What is the generic function of leukocytes? What are leukocytosis and leukopenia?	04
6.	Serotonin is a neurotransmitter which is produced by certain neurones in the brain. One of its effects is to increase the activity of sensory neurones in the brain. It also usually improves a persons mood and keeps them awake. the diagram shows a synapse at which serotonin is the neurotransmitter.  Explain how the release of the neurotransmitter serotonin, by neurone A would initiate an impulse in neurone B. Suggest one advantage of rapidly reabsorbing the serotonin.	04

7.	What are Excitatory and inhibitory postsynaptic potentials? Describe briefly.	04
8.	Briefly explain the hormonal control of digestive system.	04
SECTION – II		
Q-III Answer the following questions as directed (attempt any 10)		30
1.	Which organs can function as both an endocrine and exocrine organ? Discuss briefly.	03
2.	What are the difference between Metabolic Acidosis and Respiratory Acidosis?	03
3.	What is the intrinsic pathway for blood clotting? Discuss briefly.	03
4.	Write briefly on Sexual differentiation and sex determination.	03
5.	How are platelets formed? What is the function of platelets? What are the clinical consequences of the condition known as thrombocytopenia?	03
6.	What is the pH inside the stomach? Why is it necessary to maintain that pH level? How is it maintained? What cells produce that pH?	03
7.	What are Conventional neurotransmitters? Describe briefly.	03
8.	Write a short note on composition of Pancreatic juices.	03
9.	What are the function of liver?	03
10.	Write a short note on Organization of the Nervous System.	03
11.	What is Erythropoiesis? What are the factors needed of Erythropoiesis?	03
12.	Explain the different kinds on neurons present on the basis of their structure and function.	03
13.	Write a short note on Cystic fibrosis.	03
14.	Write a short note on Spirometer.	03